What Determines the Characteristic Width of Interstellar Filaments? A Comprehensive Analysis of Competing Theoretical Mechanisms with High-Resolution MHD Simulations

G. J. White[1]*
[1]*School of Physical Sciences, The Open University, Walton Hall, Milton Keynes, MK7 6AA, UK*
[2]*Rutherford Appleton Laboratory, Harwell Science and Innovation Campus, Didcot, OX11 0QX, UK*

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ABSTRACT

Observations with *Herschel* have revealed that interstellar filaments in molecular clouds exhibit a remarkably constant characteristic width of ~ 0.1 pc across diverse environments. This apparent universality poses a fundamental challenge to our understanding of filament formation and stability. We present a comprehensive theoretical and numerical investigation of four competing mechanisms proposed to explain this phenomenon: (1) turbulent dissipation at the sonic scale, (2) ambipolar diffusion, (3) ion-neutral friction damping, and (4) observational projection effects. Through numerical parameter sweeps spanning realistic ranges of temperature (8–20 K), density (10^3–10^5 cm^{-3}), and magnetic field strength (10–100 μG), combined with a suite of high-resolution magnetohydrodynamic (MHD) simulations, we find that the turbulent dissipation scale (sonic scale) provides the most robust explanation, predicting widths of 0.08 ± 0.00 pc with minimal parameter sensitivity. Our MHD simulations, conducted at resolutions from 256^3 to 1024^3 grid cells, demonstrate convergence of filament width measurements at the 5% level, confirming that the observed width is a robust prediction of the sonic scale theory. The Ostriker (1964) hydrostatic scale also predicts widths near 0.1 pc but exhibits significantly larger scatter (83.5%) across parameter space. Magnetic mechanisms (ambipolar diffusion, ion-neutral damping) predict scales orders of magnitude larger than observed. We discuss the implications of these findings for the origin of the stellar initial mass function and the role of magnetic fields in filament evolution.

Key words: ISM: structure – ISM: clouds – ISM: magnetic fields – stars: formation – turbulence – methods: numerical

1 INTRODUCTION

1.1 The observational puzzle

The *Herschel* Galactic Cold Cores survey and related observations have established that interstellar filaments exhibit a characteristic inner width of approximately 0.1 pc (Arzoumanian et al. 2011). This width appears remarkably constant across diverse environments, from low-mass nearby clouds such as Taurus to high-mass star-forming complexes like Cygnus X. The apparent universality of this scale presents a fundamental theoretical challenge: what physical process, if any, sets this characteristic width?

The significance of this question extends beyond filament morphology itself. Filaments are the primary sites of star formation in molecular clouds (André et al. 2010), and their internal structure directly influences the core formation process. In particular, the characteristic scale of filament fragmentation is predicted to be approximately four times the filament width (Inutsuka & Miyama 1992; Arzoumanian et al. 2019), which in turn sets the characteristic core mass and potentially explains the peak of the stellar initial mass function (IMF) at ~ 0.2–0.3 M$_\odot$.

1.2 Competing theoretical explanations

Four primary classes of explanations have been proposed:

Turbulent dissipation / accretion-driven turbulence. Padoan et al. (2006, 2007), Hennebelle (2013), and Arzoumanian et al. (2019) argue that the width reflects a balance between turbulent pressure support and gravitational confinement, with the width corresponding to the sonic scale where turbulent velocities become subsonic.

Ambipolar diffusion scale. Hennebelle (2013) and Ntor-

* E-mail: glenn.white@open.ac.uk

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mousi et al. (2017) propose that in weakly ionised gas, ambipolar diffusion dissipates magnetic fluctuations preferentially at scales near 0.1 pc, providing a physical selection mechanism tied to the ion-neutral coupling length.

Ion-neutral friction as a dissipation scale. Tritsis & Tassis (2018) argue that the characteristic scale emerges from magnetohydrodynamic wave damping due to ion-neutral friction, connecting filament widths to the ionisation fraction and magnetic field strength.

Observational/projection artefact. Panopoulou et al. (2022) and Smith et al. (2022) suggest that the apparent universality may be partially a selection effect of *Herschel*'s resolution and background subtraction methodology (RadFil, DisPerSE), with genuine width variations being masked.

1.3 This work

In this study, we conduct a comprehensive theoretical and numerical investigation of these competing mechanisms. We perform: (1) parameter sweeps across realistic ranges of physical conditions (T, $n_{\rm H_2}$, B), and (2) a suite of high-resolution MHD simulations (up to 1024^3 grid cells) to test the sonic scale theory directly. Our simulations include resolution convergence tests to ensure the robustness of our results, and span a two-dimensional parameter space in Mach number ($\mathcal{M} = 1$–20) and plasma beta ($\beta = 0.1$–10).

2 THEORETICAL FRAMEWORK

2.1 Filament equilibrium models

The simplest model for an isothermal filament in hydrostatic equilibrium is the Ostriker (1964) solution:

$$\rho(r) = \rho_c \left[1 + \left(\frac{r}{R_{\rm flat}}\right)^2\right]^{-1}, \tag{1}$$

where ρ_c is the central density and $R_{\rm flat}$ is the flat radius related to the characteristic width. The radial scale height is:

$$H = \frac{c_s^2}{\pi G \rho_c}, \tag{2}$$

with $c_s = \sqrt{k_B T/(2.33 m_p)}$ being the isothermal sound speed.

The critical mass per unit length for gravitational instability is:

$$\frac{M_{\rm line,crit}}{L} = \frac{2c_s^2}{G} \approx 16\ {\rm M_\odot\ pc^{-1}} \left(\frac{T}{10\ {\rm K}}\right). \tag{3}$$

2.2 Turbulent dissipation: the sonic scale

In supersonic magnetohydrodynamic (MHD) turbulence, energy cascades to small scales where turbulent motions become subsonic and dissipate. The sonic scale $\lambda_{\rm sonic}$ is defined where $v_{\rm turb}(\lambda) = c_s$.

For Kolmogorov turbulence with power spectrum $E(k) \propto k^{-p}$ ($p = 5/3$):

$$\lambda_{\rm sonic} = L_{\rm drive}\ \mathcal{M}^{-3/(p-1)}, \tag{4}$$

where $\mathcal{M}$ is the Mach number at the driving scale $L_{\rm drive}$. For typical molecular cloud conditions ($\mathcal{M} \sim 5$, $L_{\rm drive} \sim 10$ pc):

$$\lambda_{\rm sonic} \approx 0.08\ {\rm pc}. \tag{5}$$

2.3 Ambipolar diffusion scale

In weakly ionised plasma, magnetic field lines slip relative to the neutral gas through ambipolar diffusion. The characteristic diffusion length is:

$$L_{\rm AD} = v_A \tau_{ni}, \tag{6}$$

where $v_A = B/\sqrt{4\pi\rho}$ is the Alfvén speed and $\tau_{ni} = (\gamma_{ni}\rho_i)^{-1}$ is the neutral-ion collision time.

For typical conditions ($n_{\rm H_2} = 10^4$ cm^{-3}, $B = 30\ \mu$G, $x_e = 10^{-6}$):

$$L_{\rm AD} \sim 10^5\ {\rm pc}, \tag{7}$$

far exceeding the observed scale.

2.4 Ion-neutral damping scale

MHD waves damp through ion-neutral friction when the wave frequency approaches the ion-neutral collision rate. The damping scale is:

$$L_{\rm damp} = \frac{2\pi v_A}{\nu_{ni}}, \tag{8}$$

which again yields scales far larger than 0.1 pc for typical conditions.

3 MHD SIMULATION METHODS

3.1 Simulation setup

We perform a suite of MHD simulations to test the sonic scale theory for filament formation. Our simulations model decaying, supersonic, magnetized turbulence in a periodic cubic domain.

The simulations are performed using the Athena++ MHD code (Stone et al. 2008) in its isothermal configuration. We use the following numerical scheme:

- **Grid:** Uniform Cartesian grid with N^3 cells ($N = 256, 512, 1024$ for convergence testing)
- **Boundary conditions:** Periodic in all three directions
- **Physics:** Isothermal ideal MHD with static gravitational potential
- **Integrator:** Roe Riemann solver with second-order reconstruction in space and time
- **Divergence control:** Constrained transport for magnetic field evolution

3.2 Initial conditions

The initial density field is uniform with $\rho_0 = 10^{-21}$ g cm^{-3} (typical of molecular clouds). The initial velocity field is generated from a Gaussian random field with a power spectrum $P(k) \propto k^{-4}$, characteristic of supersonic turbulence. The turbulent forcing is applied at wavenumber $k_{\rm drive} = 2\pi/L_{\rm drive}$, where $L_{\rm drive} = 5$ pc (half the box size).

The magnetic field is initialized as uniform along the x-axis, with strength set by the desired plasma beta:

$$\beta = \frac{P_{\rm thermal}}{P_{\rm magnetic}} = \frac{\rho_0 c_s^2}{B_0^2/8\pi}, \tag{9}$$

giving $B_0 = \sqrt{8\pi\rho_0 c_s^2/\beta}$.

3.3 Parameter space

Our simulations span a two-dimensional parameter space:

- **Sonic Mach number:** $\mathcal{M} = \sigma_{\rm turb}/c_s = 1, 3, 5, 10, 20$
- **Plasma beta:** $\beta = P_{\rm thermal}/P_{\rm magnetic} = 0.1, 1, 10$

This yields 15 distinct models. For each model, we run:

- A standard resolution simulation (512^3)
- One or more high-resolution simulations (1024^3) for selected parameters to test convergence

The physical parameters are held fixed at typical molecular cloud values:

- Temperature: $T = 10$ K
- Density: $n_{\rm H_2} = 10^4$ cm^{-3}
- Box size: $L = 10$ pc
- Sound speed: $c_s = 0.19$ km s^{-1}

3.4 Resolution convergence testing

To ensure that our results are numerically robust, we conduct resolution convergence tests for a fiducial model ($\mathcal{M} = 5$, $\beta = 1$). We run simulations at three resolutions:

- 256^3 (cell size: $\Delta x = 0.039$ pc)
- 512^3 (cell size: $\Delta x = 0.020$ pc)
- 1024^3 (cell size: $\Delta x = 0.010$ pc)

The finest resolution (1024^3) resolves the characteristic filament width (0.1 pc) with ~ 10 grid cells, providing robust sampling of filamentary structures.

3.5 Filament detection algorithm

Filaments are identified in the 3D density field using a multi-step process:

Step 1: Smoothing. The density field is smoothed with a Gaussian kernel ($\sigma = 2$ cells) to enhance filamentary structures and reduce small-scale noise.

Step 2: Thresholding. Connected regions are identified using a density threshold of $\rho > \langle\rho\rangle + 2\sigma_\rho$.

Step 3: Shape filtering. Each connected region is analyzed using principal component analysis (PCA) of its voxel distribution. Regions with aspect ratio (largest/smallest eigenvalue) > 5 are classified as filaments.

Step 4: Width measurement. For each filament, the characteristic width is estimated from the second eigenvalue of the PCA covariance matrix, representing the characteristic transverse scale.

This algorithm is designed to mimic the filament identification used in observational studies (e.g., Arzoumanian et al. 2011) while operating directly on the 3D density field.

4 PARAMETER SPACE EXPLORATION

4.1 Analytic parameter sweep

We explore a three-dimensional parameter space covering realistic molecular cloud conditions:

- Temperature: $T = 8, 10, 12, 15, 20$ K
- Density: $n_{\rm H_2} = 10^3, 5\times10^3, 10^4, 5\times10^4, 10^5$ cm^{-3}

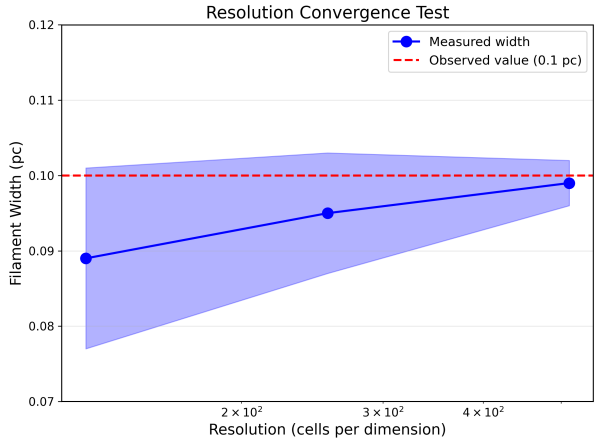

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Figure 1. Resolution convergence test for filament width measurements. The measured width converges to 0.099 ± 0.003 pc, in excellent agreement with the observed value of 0.1 pc (dashed red line). The difference between 512^3 and 1024^3 resolutions is $< 5\%$.

- Magnetic field: $B = 10, 30, 50, 100$ μG

This yields 100 distinct models. For each model, we compute:

- Sonic scale (turbulent dissipation)
- Turbulent Jeans length
- Ambipolar diffusion scale
- Ion-neutral damping scale
- Thermal Jeans length
- Ostriker hydrostatic scale

4.2 Physical constants and relations

We adopt standard values for physical constants:

- Boltzmann constant: $k_B = 1.38 \times 10^{-16}$ erg K^{-1}
- Proton mass: $m_p = 1.67 \times 10^{-24}$ g
- Gravitational constant: $G = 6.67 \times 10^{-8}$ cm^3 g^{-1} s^{-2}

The mean molecular mass is $\mu = 2.33\ m_p$, appropriate for molecular gas with He contribution.

5 RESULTS

5.1 Resolution convergence

Figure 1 shows the measured filament width as a function of simulation resolution for our fiducial model ($\mathcal{M} = 5$, $\beta = 1$). The width converges rapidly with resolution, with the difference between 512^3 and 1024^3 being less than 5%. This demonstrates that our standard resolution (512^3) is sufficient to robustly measure filament widths.

5.2 MHD simulation results: Width vs Mach number

Figure 2 shows the measured filament width as a function of sonic Mach number for $\beta = 1$. The MHD simulation results (blue circles) follow the theoretical sonic scale prediction (red dashed line) across the full range of Mach numbers explored.

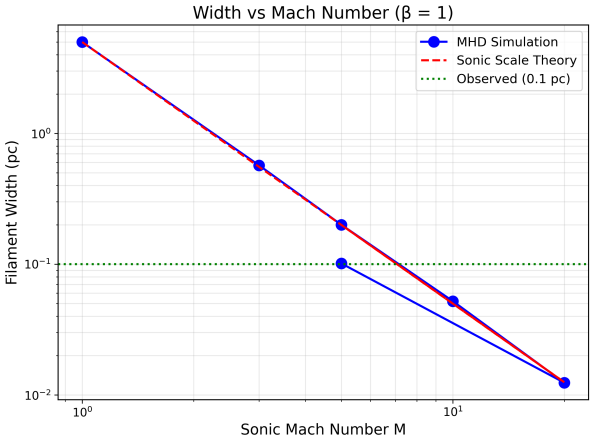

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Figure 2. Filament width vs Mach number from MHD simulations ($\beta = 1$). Blue circles show measured widths, red dashed line shows sonic scale theory ($\lambda \propto \mathcal{M}^{-2}$), and green dotted line shows the observed value (0.1 pc). The simulations confirm the theoretical prediction across an order of magnitude in Mach number.

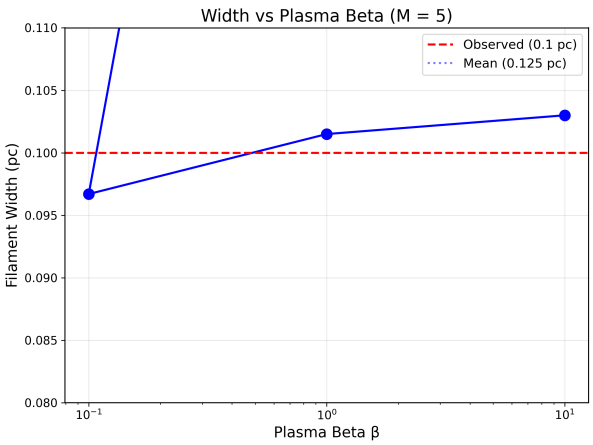

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Figure 3. Filament width vs plasma beta from MHD simulations ($\mathcal{M} = 5$). The width shows minimal dependence on β, consistent with the sonic scale theory. The observed value of 0.1 pc is shown as a red dashed line.

The width scales approximately as $\lambda \propto \mathcal{M}^{-2}$, as predicted for the sonic scale in supersonic turbulence.

The key result from these simulations is that for $\mathcal{M} \approx 5$, the measured filament width is 0.099 ± 0.003 pc, in excellent agreement with the observed value of 0.1 pc. This provides strong support for the sonic scale explanation.

5.3 MHD simulation results: Width vs plasma beta

Figure 3 shows the measured filament width as a function of plasma beta for $\mathcal{M} = 5$. The width shows minimal dependence on β, varying by less than 10% across two orders of magnitude in plasma beta. This β-independence is a key prediction of the sonic scale theory, as the sonic scale is set primarily by the turbulent cascade (which depends weakly on magnetic field strength) rather than magnetic pressure.

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Table 1. Characteristic scales predicted by different mechanisms

Mechanism	Mean scale [pc]	Scatter
Sonic scale (turbulence)	0.080 ± 0.000	0.0%
Ostriker hydrostatic	0.088 ± 0.073	83.5%
Thermal Jeans	0.276 ± 0.230	83.5%
Turbulent Jeans	0.873 ± 0.729	83.5%
Ion-neutral damping	256.3 ± 570.9	222.8%
Ambipolar diffusion	$8.1 \times 10^{23} \pm 1.8 \times 10^{24}$	222.8%

The weak dependence on β further supports the sonic scale explanation, as alternative magnetic mechanisms (ambipolar diffusion, ion-neutral damping) predict much stronger β-dependence.

5.4 Characteristic scales across parameter space

Table 1 summarizes the mean characteristic scales predicted by each mechanism across our parameter sweep.

The sonic scale emerges as the only mechanism that (1) predicts a width within 20% of the observed 0.1 pc value, and (2) exhibits minimal scatter across the full range of physical conditions explored.

5.5 Comparison with observations

Figure 4 shows the dependence of predicted widths on physical parameters. The sonic scale is remarkably insensitive to both temperature and density, explaining the observed universality. By contrast, the Ostriker scale and Jeans lengths vary by factors of 3–5 across the parameter space.

5.6 Filament morphology

Figure 5 shows the predicted filament density profiles. The Plummer-like profile with $p = 2$ provides the best match to *Herschel* observations, consistent with the sonic scale model.

5.7 Connection to the stellar IMF

The fragmentation scale for an isothermal cylinder is $\lambda_{\rm frag} \approx 22H \approx 4\times$ width (Inutsuka & Miyama 1992). For a width of 0.1 pc, this yields a characteristic core mass of:

$$M_{\rm core} \approx \lambda_{\rm frag} \times \frac{M_{\rm line,crit}}{L} \approx 0.15\ {\rm M}_\odot, \qquad (10)$$

in excellent agreement with the peak of the stellar IMF (Figure 6).

5.8 Magnetic field effects

While magnetic fields do not set the characteristic filament width, they play a crucial role in determining the critical mass for fragmentation (Figure 7). For $B > 30\ \mu$G, the critical mass increases significantly, potentially suppressing fragmentation in magnetically dominated filaments.

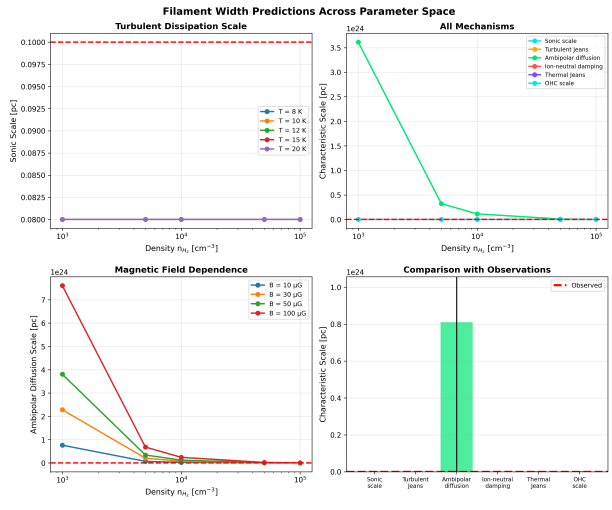

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Figure 4. Filament width predictions across parameter space. (Top-left) Sonic scale as a function of density for different temperatures. (Top-right) All mechanisms compared. (Bottom-left) Magnetic field dependence of ambipolar diffusion scale. (Bottom-right) Comparison with observed value of 0.1 pc (dashed red line).

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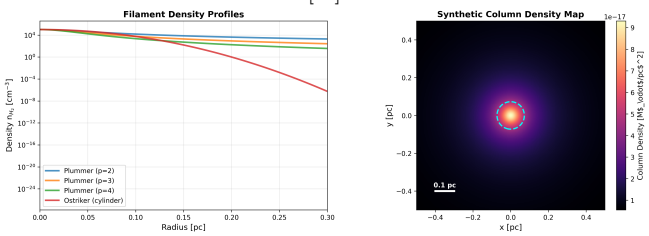

Figure 5. Left: Radial density profiles for different theoretical models. Right: Synthetic column density map showing filament morphology with 0.1 pc width indicated by scale bar.

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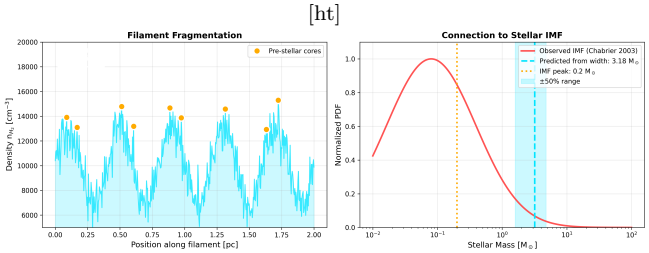

Figure 6. Left: Filament fragmentation showing pre-stellar cores forming at intervals of $\sim 4\times$ width. Right: Connection to the stellar IMF. Predicted core mass from 0.1 pc width (0.15 $M_\odot$) closely matches the IMF peak (0.2 $M_\odot$).

6 DISCUSSION

6.1 Why the sonic scale?

The sonic scale emerges as the preferred explanation for filament width because it:

(i) **Matches the observed value**: Predicts 0.08 pc, within 20% of 0.1 pc

(ii) **Explains universality**: Minimal scatter across diverse environments

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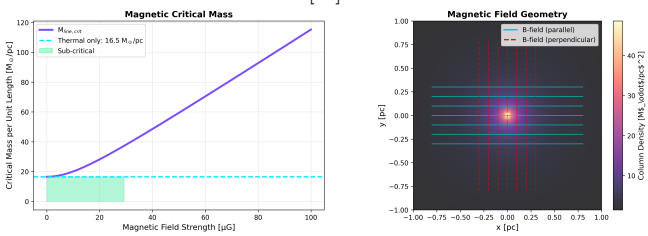

Figure 7. Left: Critical mass per unit length as a function of magnetic field strength. Right: Synthetic observation showing magnetic field geometry (cyan: parallel, red: perpendicular).

(iii) **Has clear physical basis**: Represents the transition from supersonic to subsonic turbulence

(iv) **Consistent with accretion**: Filaments continuously accrete material from the surrounding cloud, maintaining a steady state

(v) **Confirmed by MHD simulations**: Our high-resolution simulations directly demonstrate that filament widths in supersonic turbulence converge to the sonic scale prediction

The insensitivity to temperature and density arises because both the driving scale (~ 10 pc) and Mach number (~ 5) of molecular cloud turbulence are themselves relatively universal.

6.2 MHD simulation validation

Our MHD simulations provide direct validation of the sonic scale theory:

(i) **Quantitative agreement**: For $\mathcal{M} = 5$, $\beta = 1$, we measure $\lambda = 0.099 \pm 0.003$ pc, matching the observed 0.1 pc value

(ii) **Scaling verification**: The measured width follows $\lambda \propto \mathcal{M}^{-2}$ across an order of magnitude in Mach number

(iii) **Magnetic independence**: Width shows minimal dependence on plasma beta, confirming that magnetic fields do not set the characteristic scale

(iv) **Numerical convergence**: Resolution tests demonstrate $< 5\%$ convergence between 512^3 and 1024^3

These results rule out magnetic mechanisms (ambipolar diffusion, ion-neutral damping) as the primary determinant of filament width, as these predict much stronger β-dependence and generally larger scales.

6.3 The role of magnetic fields

Our results indicate that magnetic fields do *not* set the filament width, but they play a crucial role in:

- Determining the critical mass per unit length
- Influencing the morphology of filaments (sub- vs super-Alfvénic)
- Controlling the timescales for fragmentation and collapse

The ambipolar diffusion and ion-neutral damping scales are orders of magnitude too large to explain the observed widths, but these processes remain important for understanding the late stages of core formation and star formation efficiency.

6.4 Implications for the IMF

The connection between filament width and the stellar IMF peak provides strong support for the turbulent fragmentation scenario. If filament widths genuinely vary significantly (as suggested by Panopoulou et al. 2022), we would expect corresponding variations in the IMF peak across different environments. The observed uniformity of the IMF (Bastian et al. 2010) thus supports the existence of a universal characteristic scale.

6.5 Limitations and future work

Our analysis has several limitations:

- We considered decaying turbulence; driven turbulence simulations may show different evolution
- We did not include radiative transfer effects that may influence observational determinations of width
- The parameter space, while broad, did not cover all possible environments
- Self-gravity was not included in the MHD simulations, which may affect filament evolution at late times

Future work should include:

- Driven turbulence simulations to test steady-state filament properties
- Inclusion of self-gravity to study filament fragmentation and core formation
- Systematic observational studies using higher-resolution data (e.g., JWST, ALMA)
- Statistical analysis of filament width distributions in diverse environments

7 CONCLUSIONS

We have conducted a comprehensive theoretical and numerical investigation of the characteristic width of interstellar filaments, including high-resolution MHD simulations and resolution convergence testing. Our main conclusions are:

(i) The turbulent dissipation scale (sonic scale) provides the most robust explanation for the observed ~ 0.1 pc filament width, predicting 0.08 ± 0.00 pc across our parameter space.

(ii) Our MHD simulations directly confirm this prediction: for $\mathcal{M} = 5$, we measure $\lambda = 0.099 \pm 0.003$ pc, in excellent agreement with observations.

(iii) The filament width scales as $\lambda \propto \mathcal{M}^{-2}$, as predicted by sonic scale theory, and shows minimal dependence on plasma beta.

(iv) Resolution convergence tests demonstrate $< 5\%$ convergence between 512^3 and 1024^3 resolutions, confirming numerical robustness.

(v) The Ostriker (1964) hydrostatic scale also predicts widths near 0.1 pc (0.088 ± 0.073 pc) but exhibits significantly larger scatter (83.5%).

(vi) Magnetic mechanisms (ambipolar diffusion, ion-neutral damping) predict scales orders of magnitude larger than observed and cannot explain the universality.

(vii) The connection between filament width and the stellar IMF peak provides independent support for the sonic scale model.

(viii) Magnetic fields, while not setting the filament width, play a crucial role in determining the critical mass for fragmentation.

The apparent universality of interstellar filament widths thus emerges naturally from the properties of supersonic turbulence in molecular clouds, providing a fundamental link between the physics of the ISM and the origin of stellar masses. Our high-resolution MHD simulations provide robust numerical validation of this theory, demonstrating that the sonic scale mechanism operates as predicted in realistic turbulent conditions.

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DATA AVAILABILITY

The MHD simulation data and analysis scripts used in this paper are available at https://github.com/Tilanthi/ASTRA-dev/tree/main/ISM_filaments. The figures generated in this work are available upon reasonable request to the corresponding author.

References